

Responsible GeoAI: Thoughtful Integration of GIScience and AI in the Real World

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The Need for Responsible GeoAI

Recent breakthroughs in deep learning, foundation models, as well as the increasing amount of multimodal geospatial big data, have rapidly expanded the capacity and applications of GeoAI in generating geospatial knowledge, reasoning geographical processes, and supporting location-based planning and management. As GeoAI evolves from research to real-world deployment, its outputs are increasingly shaping decisions that affect people, infrastructure, and environments.

This transition raises challenges that cannot be addressed through performance improvements alone. First, GeoAI operates on inherently sensitive data. Location and location-based information can reveal individual or group behaviors, and their socioeconomic-cultural characteristics and underlying vulnerabilities. As GeoAI models infer increasingly fine-grained spatial knowledge, risks related to privacy and secondary misuse of spatial intelligence are amplified. Second, increased predictive accuracy does not guarantee that GeoAI outputs are trustworthy or appropriate for decision-making. GeoAI models may perform satisfactorily while producing overconfident or context-insensitive predictions when applied across diverse geographic, social, and environmental conditions. Third, GeoAI, if trained or used inappropriately, may reinforce spatial inequities. Models trained on data-rich regions or dominant populations often underperform in marginalized, rural, or data-scarce areas, while spatial aggregation and over-averaging can obscure geographically uneven patterns and processes. Addressing these limitations requires more than larger models, richer data, or accuracy improvements. Instead, we need a paradigm for practicing GeoAI thoughtfully in the real world.

We propose that GeoAI research and practice demand a paradigm shift toward *Responsible GeoAI*, in which ethical, theoretical, and societal considerations are embedded directly into model design, evaluation, and deployment. Although existing discussions of Responsible AI provide important guidance, they are often insufficiently place-aware (Figure 1). The real world is geospatial and introduces unique challenges related to scale, spatial aggregation, and contextual interpretation. To fill this gap, this paper conceptualizes Responsible GeoAI as a framework consisting of four interrelated dimensions: Security, Trustworthiness, Equity, and Philanthropy (STEP). It explores who can leverage GeoAI-derived spatial intelligence, when its outputs should be trusted, where the benefits and burdens of GeoAI are distributed, and how GeoAI can be employed to serve the public good.

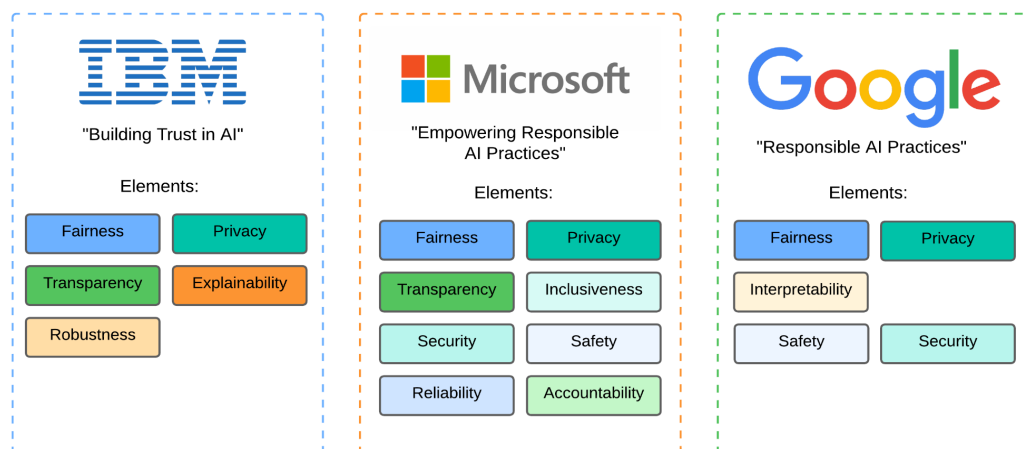


Figure 1. Components of Responsible AI Practice of IBM, Microsoft, and Google

Defining Responsible GeoAI

Responsible GeoAI refers to the design, deployment, interpretation, and governance of GeoAI in ways that thoughtfully consider spatial context, human vulnerability, and societal consequences to ensure safe, trustworthy, equitable, and publicly beneficial data, knowledge, and decisions.

Responsible GeoAI is distinct from generic Responsible AI frameworks. First, responsibility in GeoAI is inherently contextual and requires sensitivity to place. The meaning, reliability, and consequences of GeoAI outputs depend on geographic scale, environmental conditions, and social context. A model that performs well in one region or population may fail or mislead in another. Second, Responsible GeoAI is decision oriented. The responsibility of GeoAI includes not only acceptable model accuracy, but also uncertainty quantification and communication, interpretability, and guidance on appropriate use. A technically optimized model can produce harmful outcomes if its outputs are misinterpreted or over-trusted. Third, Responsible GeoAI is integrative. It requires the thoughtful integration of GIScience principles, AI methodologies, domain expertise, and stakeholder knowledge. GIScience contributes insights into geospatial scale, correlation, heterogeneity, similarity, and place-based reasoning, while AI contributes computational power, predictive and generative capacities, and pattern discovery.

The STEP Framework for Responsible GeoAI

This paper introduces the STEP (Security, Trustworthiness, Equity, and Philanthropy) paradigm as an framework for practicing Responsible GeoAI in real-world applications and elaborate on each dimension.

Security concerns the protection, governance, and controlled dissemination of geospatial data, GeoAI models, and the spatial knowledge they generate. In GeoAI, security extends beyond traditional data protection to include stewardship of inferred spatial intelligence. For example, GeoAI-based analyses of disaster victims needing rescue and evacuation can uncover highly sensitive information about individual locations or community conditions. While such timely information is valuable for life-saving emergency response, they may be misused for discriminatory or harmful practices. Responsible GeoAI therefore requires purpose-aware access control, protection, and governance mechanisms that define who can access GeoAI-derived spatial intelligence, under what conditions, and for what purposes.

Trustworthiness addresses whether GeoAI outputs can be appropriately relied upon for scientific inference and decision-making. The high accuracies brought by GeoAI may mask deeper issues related to uncertainty, calibration, and context sensitivity. For instance, urban flood simulations or damage assessments may produce visually compelling outputs that appear precise, even when uncertainty is substantial or model assumptions are violated. Responsible GeoAI emphasizes evaluation practices that extend beyond accuracy to include uncertainty quantification, robustness testing across geographic contexts, and transparent communication of model limitations. Trustworthy GeoAI should signal not only what is predicted, but also when predictions may be unreliable or inappropriate for action.

Equity focuses on the fair distribution of both the benefits and burdens of GeoAI across populations and places. GeoAI models trained on spatially uneven data may perform well in well-represented regions while failing in marginalized or under-resourced areas. Moreover, aggregate spatial outputs can obscure local disparities, masking extreme but existing geographical phenomena (e.g., neighborhoods with abnormally high exposure to air pollution). Responsible GeoAI requires equity-aware evaluation that explicitly examines where and for whom models perform poorly, as well as design strategies that mitigate bias and uneven transferability. Equity also encompasses access to GeoAI capabilities themselves. The technical and computational demands of GeoAI can concentrate analytical power within well-resourced institutions, limiting participation by smaller agencies or community organizations. Responsible practice therefore includes efforts to lower barriers through shared infrastructure, open tools, and capacity-building initiatives.

Philanthropy emphasizes alignment of GeoAI innovation with interdisciplinary knowledge discovery, public good, societal well-being. Beyond minimizing harm, philanthropic GeoAI is purpose-driven, seeking to connect technical advances with other disciplines and meaningful human and environmental benefits. For example, our research leveraged GeoAI to enable population-scale analysis of sleep behaviors by integrating geospatial data and GeoAI algorithms to explore the relationship between socio-environmental conditions and human sleep rhythms. This work expands health research beyond laboratory settings, illustrating how GeoAI can bridge experimental biology and place-based understanding to inform healthier environments.

Practicing Responsible GeoAI in Real World

Implementing Responsible GeoAI requires translating the STEP framework into concrete design, governance, and community practices. At the technical level, GeoAI should incorporate uncertainty-aware evaluation, bias-sensitive validation across places and populations, and safeguards against misuse of sensitive spatial data and models. At the governance level, role- and purpose-based access to GeoAI models and outputs can balance the need for actionable insight with privacy and security considerations. At the community level, advancing Responsible GeoAI depends on interdisciplinary collaboration, shared infrastructure, open educational resources, and engagement with stakeholders whose living environments and behaviors are shaped by GeoAI-informed decisions. Practically, a checklist can be developed to determine and evaluate the responsibility of a GeoAI model or research. As GeoAI increase to develop and play critical roles in the real world, future work may explore fiduciary GeoAI perspectives that move beyond best practices toward explicit duty-based obligations to the people and places impacted by GeoAI-informed decisions.